Venkata Lakshmi Parimala Pasupuleti

Senior Python Developer · Income Verification & Credit Analytics · ML Model Implementation · Financial Data Pipelines · AWS · Full Lifecycle · 9yr Production

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PROFESSIONAL SUMMARY

Senior developer with 9+ years building production analytics software on regulated financial data -- specializing in automating income eligibility, asset verification, and employment assessment workflows at enterprise scale. At Entrata, I owned the income and asset verification pipeline for federally funded housing programs for 7 years: 1.6M HUD/LIHTC/HCV income-limit records normalized annually, AMI threshold automation, recertification workflows, and borrower-capacity calculations for 100K+ housing units -- with zero compliance violations across the full ownership period. This is the same analytical problem CIA automates for mortgage underwriting, applied to the rental side of the housing market. I bring strong Python engineering (FastAPI, Flask, full lifecycle), ML model implementation (Random Forest, Logistic Regression, Isolation Forest -- 85%+ accuracy on 104K-record datasets), SQL optimization (200+ queries, 30% performance gain), and automated testing infrastructure across production systems. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr production), SQL/PL-SQL, R, Java, JavaScript, TypeScript, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Django (familiarity), Spring Boot, Angular, React.js

Analytics, ML & Data Engineering: Machine Learning (Random Forest, Logistic Regression, XGBoost, Isolation Forest, BERT), Model Implementation & Evaluation, Feature Engineering, Statistical Modeling, Anomaly Detection, Data Analysis (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow), Jupyter, Tableau, Power BI, ggplot2, ETL/ELT Pipelines, Financial Data Pipelines, Big Data Processing, Income Verification Automation, Data Modeling, Data Normalization, RDBMS (PostgreSQL, MySQL, Oracle, MS SQL Server), NoSQL, SQLite

Development Tools & Infrastructure: Git, Make, shell scripts, pdb (Python debugger), GitHub Actions CI/CD, Automated Testing, Regression Testing, Unit Testing, Docker, Kubernetes, AWS (S3, EC2, RDS, Glue), AWS EKS, Azure (App Service, SQL), GCP, RabbitMQ async processing, Microservices, REST APIs, Performance Optimization, Observability, Structured Logging, Agile/Scrum, Technical Mentoring, Code Reviews

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Income Verification, Credit Analytics & Data Pipelines · Lehi, UT (Remote)

- **Income & asset verification automation -- 7-year ownership** -- sole owner of Entrata's HUD/LIHTC/HCV income-limit import and eligibility system: annual ingestion of federal income-limit data (few thousand source rows expanded to ~180,000 domain records), AMI threshold automation, household income certification, asset documentation, and recertification workflows for 100K+ housing units. Every release required pre-import baseline validation, reconciliation counts, and rollback preparation -- a wrong import triggers incorrect eligibility decisions for thousands of residents. Zero compliance violations across full ownership period. Direct analog to CIA's borrower capacity automation for mortgage underwriting.
- **Analytics software & model implementation** -- built income fraud detection logic (AMI cutoff rules, household composition validation, income-source cross-checks) embedded in eligibility calculations; designed stored procedures and set-based transformations for large-scale financial record processing; optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, and partitioning; automated 80% of manual data operations via Python/PL-SQL pipelines.
- **Full life cycle production delivery (full stack)** -- shipped 20+ full-stack modules (Python, JavaScript, PostgreSQL) across 120+ Agile sprints; built and maintained robust, maintainable code with comprehensive documentation and knowledge-sharing resources; RabbitMQ async processing with retry logic and idempotency guarantees; Docker microservices redesign (50% scalability improvement); CI/CD with automated regression testing; mentored 5+ engineers; 300+ peer code reviews.
- **Cloud & enterprise integration** -- led AWS S3 document migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); gathered and translated business requirements into technical specifications; integrated enterprise workflows with REST APIs, third-party data providers, and multi-tenant RDBMS systems across complex financial and compliance schemas.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) on WHO and UK Road Safety datasets across 190+ countries (R2=0.977 regression accuracy); applied Isolation Forest to automate data integrity checks across 3 research pipelines, surfacing anomalies that manual review missed -- equivalent to income fraud signal detection; designed Tableau, ggplot2, and Quarto dashboards improving research comprehension 40% for faculty leadership -- present results to non-technical audiences as part of standard research workflow; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Analytics Pipeline & Automated Testing

- Built end-to-end analytics software: business rules implementation in Python, automated validation pipeline with compliance guardrails, CI-integrated testing gate, and observability stack -- sole technical owner from specification through production delivery.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Data Pipelines

- Built income-limit import pipeline for 1.6M HUD/Tax Credit records across 5 years; property management and compliance modules (Python, PostgreSQL); AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification document templates.

PROJECTS

CrashClarity: Large-Scale Predictive Analytics -- 104K Records, ML Model Implementation

Machine Learning · Python · R · scikit-learn · Model Implementation · Analytics · GitHub Pages

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Pandas · Matplotlib · GitHub Pages | venkatalakshmiparimala.github.io/Portfolio/project_crashclarity.html

End-to-end predictive analytics implementation on 104K+ real-world records -- demonstrates the full model development lifecycle CIA uses for credit analytics. **Model implementation:** Random Forest + Logistic Regression pipeline with engineered features across weather conditions, road types, speed limits, time-of-day variables, and severity labels; 85%+ model accuracy on held-out test set. **Analysis pipeline:** full lifecycle from raw data ingestion and cleaning through feature engineering, model training, evaluation, and results presentation; hyperparameter tuning with cross-validation; performance comparison across model families. **Reporting:** interactive analytical dashboard deployed on GitHub Pages -- visualizing variable importance, prediction distributions, and accuracy metrics for non-technical stakeholders. Demonstrates the same analytical software design, model implementation, and result presentation skills CIA applies to borrower capacity assessment.

Leasel: Income & Asset Verification AI Platform -- Compliance Automation

Income Verification · Compliance Analytics · Python · FastAPI · Automated Testing · Regulated Data

Python · FastAPI · LangGraph · TF-IDF · SQLite · REST APIs · pdfplumber · OCR | github.com/venkatalakshmiparimala/Lease_complaince_rag

AI-powered income verification and compliance automation platform -- the software analog of what CIA builds for mortgage borrower assessment, applied to residential lease compliance. **Business rules implementation:** automated income eligibility checks, asset documentation verification, and household certification workflows across LIHTC/HCV/HUD regulatory standards -- 93.6% precision, 87.9% F1 across 1,240 compliance test cases on residential lease PDFs (5 US states + federal, 1,050 synthetic tenants). **Testing infrastructure:** automated test harness with 225-query benchmark dataset, regression gate blocking deploys on score degradation, multi-trial sampling to detect non-deterministic variance. **Core design rule enforced in code:** analytics software extracts and reasons; human or deterministic logic makes every final regulated compliance call -- no AI auto-resolution of eligibility or financial decisions.

AgentProbe: Automated Testing Framework & Analytics Platform

Automated Testing · CI/CD · FastAPI · Python · Analytics · Open Source

Python · FastAPI · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic / OpenAI / Groq APIs | github.com/venkatalakshmiparimala/AgentProbe

Production automated testing and analytics framework -- CI-integrated quality gate that validates analytical software output before each deployment. **Testing infrastructure:** 5-metric automated evaluation engine (correctness, completeness, accuracy, hallucination, latency), CI pipeline (GitHub Actions) blocking quality degradation, multi-trial regression detection across test runs. **Analytics platform:** 15-endpoint FastAPI backend, REST API design, structured observability and result tracking across all evaluation runs. Demonstrates the automated testing strategies, scalable software architecture, and quality-gate discipline CIA requires for reliable credit analytics software.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017